Tuesday Warm-up: Motional emf and Faraday's Law

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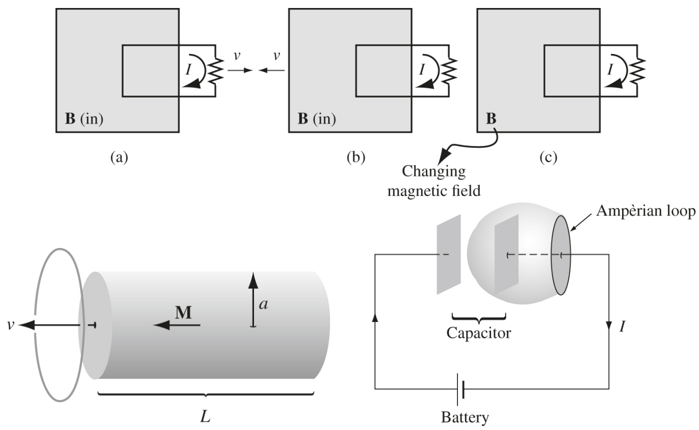

Figure 1: (Top) Three tests of Faraday's Law. (Bottom left) A magnet passes through a loop of wire at constant velocity. (Bottom right) An emf charges a capacitor.

1 Memory Bank

- **Definition of emf**:

$$\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{l} \tag{1}$$

- **Flux rule and emf**:

$$\mathcal{E} = -\frac{d\Phi}{dt} \tag{2}$$

$$\Phi = \int \mathbf{B} \cdot d\mathbf{a} \tag{3}$$

- **Faraday's Law**:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{4}$$

2 Ohm's Law and Motional emf

1. Consider Fig. 1 (top). (a) Why is there an induced emf in the wire loop that moves to the left? (b) Why is there an induced emf in the wire loop if the **B**-field moves to the right? (c) Why is there an induced emf if the strength of the field in the wire loop changes?
2. Consider Fig. 1 (bottom left). A long cylindrical magnet of length L and radius a carries a uniform magnetization $\mathbf{M}$ parallel to its axis. It passes at constant velocity v through a circular wire ring of radius b. Graph the induced emf as a function of time.

3 Faraday's Law

1. Using the definition of emf and magnetic flux, construct a proof for the differential form of Faraday's law, via Stoke's Law.
2. Suppose two identical solenoids with turns per unit length n and radius a are arranged concentrically. The following emf is connected to one solenoid:

$$\mathcal{E} = \mathcal{E}_0 \sin(2\pi f t) \tag{5}$$

   What is the emf induced in the other solenoid?
3. Turns out we need to fix Ampère's Law. Consider 1 (bottom right). What is I_{enc} for the flat and curved surfaces?